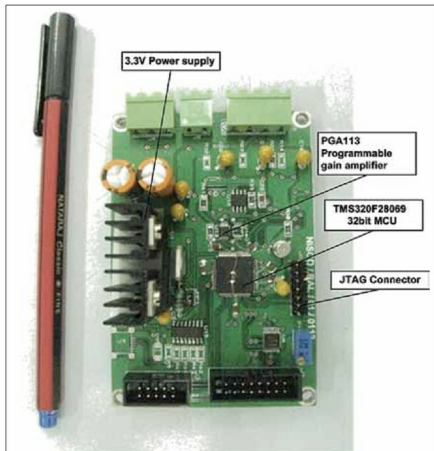




The card

TI's 32-bit Piccolo series floating-point real-time processor, TMS320F28069, which is clocked at 80 MHz. The device consumes 5W power and uses state-of-the-art digital signal processing to extract mass unbalance and its angle. The operator then uses the information to adjust the parameters of the rotating object and reduce the vibrations.

The unbalance in the rotating body is sensed by a piezo-sensor. The output of the piezo-sensor is amplified by

using PGA113, a programmable gain amplifier, and then digitised using on-chip 12-bit analogue-to-digital converter. An inductive proximity sensor is used to sense speed and to generate reference signal. By extracting the fundamental component of the vibration signal, the amount and angle of the vibration are found out.

As the system relies purely on digital signal processing, the accuracy and precision, repeatability are greatly enhanced. The resulting circuit is very

compact and much more efficient. To reduce the cost, the developers reduced the number of components and selected a fixed-point processor.

Hurdles aplenty

"Whenever you start something very new, you will face challenges. I mean, in many ways, that was one sign that we were heading in the right direction," says Mane.

He explains: "The output of the piezo-sensor is highly noise-contaminated and it is very difficult to extract the unbalance signal. This problem aggravates in low-rpm machines—300-400 rpm machines of 1-3 tonne capacity. We worked on the design for a year and after a lot of debugging and tweaking, we managed to get the design right. Thankfully, all the pieces locked into action and have worked seamlessly." The new indicator has undergone successful field trials for all machine capacities ranging from 5 kg to 3 tonnes and compliance for ISO1940.

Future plans

"The product is in the finished state and applicable for both vertical and horizontal type dynamic balancing machines. Our immediate plan is to patent this design. We are working on adding user-interface facilities like TFT display and USB interface before it is launched nation-wide. It will be available as a sub-assembly of balancing machines manufactured by Balancing Instruments and Equipments Miraj Pvt Ltd, Miraj, India," informs Mane. ●

The author is a tech correspondent at EY Bengaluru

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